Atomic elementary flux modes explain the steady state flow of metabolites in large-scale flux networks

aEFMs explain metabolite flows

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Flux network/Elementary flux mode/flow decomposition/Markov chain/atom mapping

Abstract

Steady state fluxes are a measure of cellular activity under metabolic homoeostasis, but understanding how individual substrates are metabolized remains a challenge in large-scale networks. Pathway-based approaches such as elementary flux mode (EFM) analysis are limited to small networks due to the combinatorial explosion of pathways and ambiguity decomposing fluxes onto EFMs. Here, we present an alternative approach to explain metabolic fluxes in terms of the steady state flow of their atomic constituents. We refer to these pathways as atomic EFMs (AEFMs) and show that computations involving AEFMs are orders of magnitude faster than standard EFMs which balance molecular stoichiometries. Using our approach, we enumerate carbon and nitrogen AEFMs in five genome-scale metabolic models and compute the AEFM decomposition of fluxes estimated in a HepG2 liver cancer cell line. Our results systematically characterize carbon and nitrogen remodelling and, on the HepG2 network, predict glutamine metabolism through the recently discovered non-canonical TCA cycle. We anticipate our method will be useful for quantifying cellular metabolism as high-throughput fluxomic analyses become increasingly common.

Introduction

Recent technological and computational advancements are leading to a rise in metabolic flux data (Lee et al, 2019; Wang et al, 2022a,b). These fluxes quantify the rate of metabolite interconversions within a cell or their movements across subcellular compartments. Metabolic fluxes are studied to understand the dynamics of substrate switching (Zampieri et al, 2019; Kochanowski et al, 2021), redox balancing (McKinlay & Harwood, 2010; Franchina et al, 2022), overflow metabolism (Basan et al, 2015; Vander Heiden et al, 2009), cell differentiation (Endo et al, 2022; Jackson & Finley, 2024), and cellular communities (Schuetz et al, 2012; Yu et al, 2022; Gustafsson et al, 2024). Hence, 'fluxomics' (Emwas et al, 2022; Sanford et al, 2002) is emerging as a field dedicated to quantifying the totality of fluxes within biological systems.

Current experimental and computational methods generally estimate fluxes under metabolic homeostasis. Under this assumption, the production and consumption of metabolites are balanced over time, and the fluxes are said to be under steady state. Techniques such stable isotope tracing analysis (SITA) experimentally determine steady state fluxes by culturing cells in labelled substrates and quantifying their incorporation within endogenous metabolites (Buescher et al, 2015; Antoniewicz, 2018). Using this method, fluxes have been measured in major metabolic subsystems such as glycolysis, pentose phosphate pathway, and the TCA cycle (Orth et al, 2010b; Ahn et al, 2017; Wu et al, 2019). Many computational methods have also been proposed to infer steady state fluxes from transcript, protein, and metabolite abundance data (Uematsu et al, 2022; González-Arrué et al, 2023; Huang et al, 2023; Kaste & Shachar-Hill, 2023; Lee et al, 2023). Collectively, these approaches are being used to es-

timate and predict fluxes in genome-scale metabolic models (GEMs) involving hundreds to thousands of reactions (Nilsson et al, 2020b; Cherkaoui et al, 2022; Gelbach et al, 2024).

Although much effort has been made to estimate steady state fluxes, less emphasis has been placed on developing methods to analyze these large-scale flux datasets. In contrast to abundance data (e.g. transcripts, proteins, metabolites), steady state fluxes are constrained by the reaction stoichiometries between metabolite inflows and outflows. In a network with fixed source and sink fluxes an increase in flux along one metabolic pathway must decrease fluxes along one or more other pathways to maintain mass conservation. For simple networks, one may predict how the system would evolve by identifying the up and downstream reactions flanking an enzymatic perturbation. For example, Schwartz & Kanehisa were able to enumerate all glycolytic pathway fluxes in a 12 reaction model of yeast glycolysis (Schwartz & Kanehisa, 2006). Yet in more complex genome-scale metabolic networks containing thousands of fluxes (e.g. Brunk et al (2018); Robinson et al (2020); Wang et al (2021)), there exists no hierarchical relationship between reactions. A change in one reaction rate, let alone multiple perturbations, may alter network fluxes beyond neighbouring reactions (Nobile et al, 2021).

This problem of understanding the flow of metabolites within steady state flux networks has historically been addressed through the concept of elementary flux modes (EFMs). EFMs are a pathway-based approach to analyze metabolic networks and are defined as minimal sets of biochemical reactions that maintain steady state flux (Schuster & Hilgetag, 1994). This minimal property specifies that an EFM cannot be decomposed into two or more smaller pathways carrying steady state flux. An attractive property of EFMs is that any set of steady state fluxes in a network can be explained as a positive, linear combination of EFMs (Schuster

et al, 1999). While EFMs are a mechanistic explanation of how metabolites travel through networks, the number of EFMs scales combinatorially as a function of network size (Acuña et al, 2009). It remains computationally infeasible to enumerate EFMs in GEMs after over a decade of algorithmic advancements (e.g. Gagneur & Klamt 2004; Terzer & Stelling 2008; Buchner & Zanghellini 2021). The number of EFMs also poses a challenge for explaining network fluxes in terms of their EFMs. When there are more EFMs than linearly independent fluxes, there exist multiple ways to assign EFM weights that reconstruct the observed network fluxes. Consequently, structural analyses of EFMs and downstream analyses of their weights have been limited to small-scale metabolic subnetworks (Fakih et al, 2023; Mpabanzi et al, 2022; Toya & Shimizu, 2022; Matsuda et al, 2011; Schwartz & Kanehisa, 2006).

In our previous work, we took the first steps to solve the EFM flux decomposition problem for the special case of unimolecular reaction networks under a Markov constraint. We modelled a hypothetical particle moving between metabolites under the assumption that the transition probability of each reaction was proportional to the observed steady state flux. By expanding the notion of Markov chain state to include sequences of metabolites, we proposed an algorithm to efficiently compute simple cycle probabilities corresponding to a given EFM. We called this construct the *cycle-history Markov chain* (CHMC) and proved that these EFM probabilities could be scaled to weights that fully reconstructed the flux along each reaction (Chitpin & Perkins, 2023).

In this work, we generalize our CHMC method to operate on any type of metabolic network including those involving multispecies reactions. We do so by proposing a new type of *atomic elementary flux mode* which traces the steady state flow of an individual atom within a metabolic network. We refer to these pathways as AEFMs to distinguish them

from the (molecular) EFMs proposed by Schuster & Hilgetag. While a linear combination of molecular EFMs will reconstruct the overall reaction fluxes, the totality of AEFM weights for a given source metabolite will always reconstruct its mass flow entering the network. We first introduce our framework for enumerating and uniquely identifying AEFM weights. Using a state-of-the-art atom mapping algorithm, we show how to generate atomic transition graphs from which we construct atomic cycle-history Markov chains (ACHMCs). From these ACHMCs, we enumerate AEFMs in GEMs considered infeasible by state-of-the-art molecular EFM enumeration programs. We further analyse the structure of the AEFMs to characterize nutrient source remodelling. Lastly, we analyze a single set of AEFM weights estimated from a HepG2 liver cancer cell line to quantify glutamine metabolism. Our results predict glutamine in HepG2 cells is metabolized through a non-canonical TCA pathway recently discovered in non-small cell lung cancer (NSCLC), mouse embryonic, and C2C12 mouse myoblast cells (Arnold et al, 2022).

Results

Tracing atomic flows in genome-scale models

We first describe what molecular EFMs look like in metabolic networks and provide intuition why enumerating and decomposing fluxes onto these pathways remains a fundamental problem in literature. Figure 1a shows an example molecular EFM involving substrates glucose and ATP which are consumed to produce metabolites G3P, DHAP, and ADP. Multiple source and sink fluxes are required to balance the internal metabolite stoichiometries, lead-

ing to several branching and converging paths within the molecular EFM. As the metabolic networks expands to include dozens or more reactions, the corresponding molecular EFMs can grow increasingly complex with numerous source and sink fluxes required to stoichiometrically balance participating metabolites. In particular, there may be multiple sets of reactions involved to balance pairs of energetic molecules (e.g. ATP/ADP), redox equivalents (e.g. NAD^+/NADH), metabolic byproducts (e.g. water, hydrogen ions), or coenzymes (e.g. coenzyme A). This complexity can lead to an intractable number of pathways that may only differ by reaction subsets required to mass-balance these metabolites. Choosing certain metabolites to remove may, in part, solve this problem. However, the choice of metabolites to prune from the network is often subjective and not guaranteed to sufficiently reduce the total number of molecular EFMs.

Given many metabolic flux networks are generated to study nutrient catabolism (Zamboni et al, 2015), we argue for studying source metabolite-derived AEFMs to reduce the combinatorial number of molecular EFMs. We highlight an example carbon AEFM in Figure 1b which shows how the C6 glucose carbon is remodelled during the first steps of glycolysis. As atoms are never split apart or fused together in metabolic reactions, these AEFMs always correspond to simple paths spanning source and sink metabolites or simple cycles between internal metabolites. These pathways are straightforward to interpret because each atom within an AEFM can only transit from a single substrate to product, regardless of the network size, connectivity, or number of participating substrates/products in each reaction. This property further enables us to enumerate and compute the AEFM weights using our previously described CHMC method to uniquely decompose steady state fluxes onto EFMs for unimolecular reaction networks (Chitpin & Perkins, 2023).

We outline our computational pipeline in Figure 1c to enumerate AEFMs and uniquely assign their weights when the steady state fluxes of the network are known. In step 1, we require as inputs the metabolic network defined by a set of metabolites and reactions encoded as a stoichiometry matrix, the metabolite structures denoted as SMILES strings, and the steady state fluxes. From the stoichiometry matrix and SMILES strings, we construct reaction SMILES strings which are imputed into the atom mapping program RXNMapper (Schwaller et al, 2021) to map the position of each atom within each reaction equation. This atom mapping data is used to construct an atomic transition graph in step 2, which traces the atomic position of an individual atom from a given source metabolite entering the network. We set the transition probabilities proportional to the atomic flux through each reaction, accounting for those with multiple substrate stoichiometries (see Methods and Figure EV 5). For each atomic transition graph, we construct the corresponding ACHMC to enumerate all AEFMs involving that source metabolite atom in step 3. Finally, the AEFM weights are computed in step 4 by steady state analysis of the ACHMC.

Enumerating AEFMs is computationally tractable in large-scale networks

Given the major computational challenge of enumerating molecular EFMs in networks with > 100 metabolites and reactions, we first sought to test whether it was feasible to enumerate AEFMs in GEMs. We selected five genome-scale metabolic models across distinct organisms in ascending order of network size. These networks include a metabolic model of *E. coli* (E. coli core; Orth et al (2010a)), human red blood cells (iAB RBC 283; Bordbar et al (2011)),

H. pylori (iIT341; Thiele et al (2005)), *S. aureus* (iSB619; Becker & Palsson (2005)), and human liver cancer cells (HepG2; Nilsson et al (2020a)). These models were pre-processed to ensure unambiguous atom mappings across all reactions. Namely, all metabolites with no known structures (pseudometabolites), and pseudoreactions with non-integer stoichiometries were removed from the networks. The resulting metabolic networks ranged between 71-547 metabolites and 144-974 reactions (Figure EV1). We then attempted to enumerate the molecular EFMs using FluxModeCalculator, which is the fastest molecular EFM enumeration program (van Klinken & Willems van Dijk, 2016). Following our pipeline, we subsequently compiled the metabolite SMILES strings and constructed ACHMCs rooted on each carbon and nitrogen atom across all source metabolites. The resulting summary statistics describing the GEMs and ACHMCs are presented in Table 1. Across all carbon ACHMCs, we observed a broad increase in the total number of AEFMs as a function of metabolic network size. The one exception to this observation was the HepG2 network which contained fewer metabolites and reactions than iSB619, yet exhibited two orders of magnitude more carbon AEFMs. Inspection of the HepG2 network revealed it only contained 40 source metabolites compared to the 198 source metabolites in the iSB619 network, indicating a greater degree of network connectivity between internal metabolites in the HepG2 network (Figure EV1).

Figure 2a shows the total number of atomic and molecular EFMs enumerated by our program MarkovWeightedEFMs.jl and FluxModeCalculator on the five GEMs. Using our ACHMC pipeline, we enumerated between $10^3 - 10^6$ carbon and $10^2 - 10^3$ nitrogen AEFMs within each GEM. In contrast, FluxModeCalculator failed to enumerate molecular EFMs in all but the two smallest E. coli core and iAB RBC 283 datasets, returning $10^3 - 10^4$ times more molecular-versus-atomic EFMs.

By focusing on an atom of interest, our ACHMC approach greatly reduced the number of EFMs to biologically relevant pathways involving nutrient source remodelling. This reduction in pathways resulted in AEFM-versus-EFM enumeration completing $10^3 - 10^4$ times faster (Figure 2b). While running FluxModeCalculator in parallel did improve run times by 8.2-8.6 times (Figure EV2), it remained over $10^1 - 10^2$ slower than enumerating their corresponding carbon and nitrogen AEFMs. We further found that AEFM enumeration actually involved fewer metabolite-atom combinations than the total number of metabolites in a given network. Constrained by the atom mapping predictions, we observed that the average carbon and nitrogen ACHMC across all five GEMs only transited 0.41% and 1.7% of the total number of metabolite-carbon and metabolite-nitrogen positions, respectively (Figure EV3). While serial run times in Figure 2c scaled quadratically (Figure 2c; slope $= 1.90$, intercept $= -6.54$), our program and AEFM approach benefits greatly from parallelism. Each ACHMC can be computed independently of one another and our program is also parallized on the level of individual ACHMCs to improve AEFM enumeration. For the top 28 largest carbon ACHMCs in the HepG2 network, Figure 2d shows a median speedup of 7.1 times when using 32 threads.

Most AEFMs are source-to-sink pathways spanning dozens of reactions

We next turned to characterizing the structural properties of all carbon and nitrogen AEFM across the five GEMs. Molecular EFMs can be divided into pathway that span source and sink metabolites or internal loops within the metabolic network such as pairs of reversible reactions. Both pathways have clear biological interpretations with source-to-sink pathways

explaining the steady state flow of source metabolites on their way to exiting the network, and internally looped pathways corresponding to substrate cycles. At an atomic level, these pathways can be subclassified further into AEFMs that transit sets of distinct metabolites or those that revisit the same metabolite, albeit in different atomic positions. This notion of a "metabolite revisitation" is equivalent to an isotopomer in stable isotope tracing (Antoniewicz, 2018). An example source-to-sink AEFM with this property is a glucose-derived carbon AEFM remodelling through two turns of the TCA cycle; on the second turn, the position of the carbon atom shifts within the TCA intermediates, resulting in different isotopomers (Duan et al, 2022) and therefore a source-to-sink pathway with a metabolite revisitation under our nomenclature.

To obtain a broad understanding of the types of AEFMs within the five GEMs, we categorized them into source-to-sink pathways and looped pathways with or without metabolite revisitations in different atomic positions. Figure 3a-b shows the counts for each AEFM classes within each GEM. As expected, we observed that the majority of carbon AEFMs were source-to-sink pathways with or without metabolite revisitations. Those without metabolite revisitations tended to scale as a function of network size, while those with revisitations increased modestly. Investigation of these source-to-sink pathways confirmed that many metabolite revisitations involved TCA cycle intermediates. Looped AEFMs without metabolite revisitations primarily corresponded to transport reactions cycling metabolites across compartments and cyclic pathways involving TCA intermediates. These two observations may explain why the iAB RBC 283 network contained the fewest number of pathways with metabolite revisitations, since they lack organelles and cannot support cyclic AEFMs associated with metabolite transport between compartments nor mitochondrial TCA activity.

There were very few internally looped AEFMs with metabolite revisitations with no obvious trend across the five GEMs. The greatest number of these pathways were found in the E. coli core network with many metabolite revisitations involving carbon dioxide derived from isocitrate and alpha-ketoglutarate re-entering the TCA cycle via oxaloacetate. As there are fewer metabolites containing nitrogen, we found the majority of their AEFMs corresponded to pathways without metabolite revisitations, primarily involving amino acid remodelling or transports between compartments. Looped pathways with metabolite revisitations were only observed in the iIT341 and iSB619 networks through reactions involving the production and consumption of ammonia.

Since many looped AEFMs involve reversible reactions or metabolite transport across compartments, we hypothesized that looped AEFMs should involve fewer reactions than source-to-sink pathways which can span numerous internal metabolites. We calculated the fraction of carbon and nitrogen AEFMs with given lengths across all five GEMs and plotted their distributions in Figure 3c-d. Our results show that carbon AEFMs in the first four networks are bimodally distributed with the majority of pathways involving a dozen or fewer reactions. While carbon AEFMs of the HepG2 network appeared to exhibit a single Gaussian distribution ($\mu = 61.7$, $\sigma^2 = 15.5$), closer inspection of the histogram revealed a small peak spanning 2-12 reactions. We considered whether these two peaks observed in all five GEMs were associated with source-to-sink and looped AEFM classes. We tested this hypothesis by reclassifying the carbon AEFMs spanning less than 5-12 reactions depending on the GEM (shorter pathways), and the remaining carbon AEFMs (longer pathways). We found that long pathways were associated with source-to-sink pathways, with shorter pathways containing the majority of looped pathways. In fact, 65-95% of all longer pathways

in each GEM were source-to-sink pathways with metabolite revisitations, with this statistic increasing to 95-100% when including source-to-sink pathways without metabolite revisitations. Among shorter pathways, we observed that roughly half or more of them were looped carbon AEFMs with or without metabolite revisitations. Analysis of the nitrogen AEFMs revealed only a single distribution of relatively short pathways, since there are much fewer nitrogen-containing metabolites across all GEMs.

We conclude our structural analysis by examining the AEFMs accessible to distinct atoms within the same source metabolite, irrespective of their class. We first computed the number of AEFMs for each carbon atom (resp. nitrogen atom). For each metabolite, we then computed the ratio of the maximum number of paths for any carbon (resp. nitrogen), and the minimum number of paths. So for example, a ratio of 10 means that one of the carbon atoms can transit the network in 10 times as many different ways as the carbon with the fewest paths through the network. Our results are shown in Figure 3e-f, where each dot corresponds to the ratio for a given source metabolite. We found that a surprising number of carbons and nitrogens can transit different pathways within their respective source metabolites. Focusing on the source metabolites with carbon and nitrogen ratios over 100, we observed many of these corresponded to hydrophobic and positively-charged amino acids such as phenlylalanine, methionine, arginine, and histidine. Source metabolites containing fewer carbons or nitrogens tended to exhibit ratios closer to 1 with AEFMs involving the same metabolite subsets. Metabolites such as glycine, for example, are equivalently metabolized across the same carbon AEFMs within each GEM. For each GEM, and not counting source metabolites with ratios of one, we computed median carbon ratios between 2.0 and 46.8 and nitrogen ratios between 1.3 and 11.2. Overall, these findings highlight the varying degrees of

metabolic flexibility for atomic constituents of source metabolites to transit the metabolic network.

A minority of carbon AEFMs explains the majority of [illegible]bolite remodelling

Explain in 2-3 sentences how fluxes measured / estimated.

We finally highlight the potential of our AEFM method for underst[illegible]rce metabolism. Specifically, we analyzed the single set of steady state fluxes from the HepG2 network to identify alternative pathways of glutamine metabolism. This question is highly relevant since many cancer cells, including HepG2, are dependend on this non-essential amino acid for survival and proliferation (Ye et al, 2023; Yoo et al, 2020). Using our method, we decomposed the set of steady state fluxes onto the 241,085 glutamine carbon AEFMs characterizing the HepG2 metabolic network. Figure 4a shows the cumulative explained mass flow of each glutamine carbon as a function of their AEFMs ranked from greatest to smallest explained mass flow. We capped the cumulative explained mass flow of each glutamine carbon at 99% because we observed that 98.9% of AEFMs collectively explained the remaining 1% glutamine carbon flow. These results suggested that the majority of glutamine carbon remodelling is explained by a surprisingly small number of glutamine AEFMs, especially since our Markovian flux decomposition approach assigns some weight to all network pathways. The top 5 ranked AEFMs across each glutamine carbon (25 pathways altogether) explained over half (50.2%) of the total glutamine carbon mass flow with the subsequent 5 next highest ranked AEFMs only explaining an additional 9.8% glutamine carbon mass flow. These observations were consistent across carbon AEFMs derived from

EV4

other amino acids as well as glucose, leading us to hypothesize that stea[illegible] flu[illegible] a relatively small subset of AEFMs (Figure EV3). For some source [illegible] verlapping curves between carbons (e.g. arginine, glycine, serine in Fi[illegible] that individual carbons were remodelled through the same AEFMs. However,s

Corroborating results from Figure 3e-f, we observed many non-overlapping curves for the majority of amino acids since their carbons could transit the network via [illegible] metabolites.

To obtain a high-level understanding of the dominant pathways of [illegible] kill based on ... predictions

modelling, we constructed a metabolic subnetwork from the top 5 glutam[illegible] collapsing metabolite-atom positions onto distinct metabolites. Of th[illegible] the HepG2 dataset, input glutamine can remodel into 202 species, based on the RXNMapper atom mapping predictions, localized to t[illegible]dria. We found that the top 5 glutamine-derived carbon AEFMs con[illegible]es and 31 reactions (Figure 4b). These AEFMs generally consisted [illegible] involving TCA intermediates. Only two source-to-sink reactions converting cytosolic glutamine towards the protein pool and carbon dioxide through alpha-ketoglutarate decarboxylation in the mitochondria.

convert

As AEFMs are steady state pathways, they can be visualized as [illegible]ter understand the carbon flows through the network. Figure 4c-e s[illegible] for each glutamine carbon. Each AEFM in the diagram is shown [illegible]ional to its AEFM weight with nodes representing the metabolite transited by the given glutamine carbon. Although each glutamine carbon can transit a different number of AEFMs, we found the top 5 backbone carbons and first two sidechain carbons were identical in both AEFM pathway and weight.

transitted

carbon

Focusing on either backbone carbons in Figure 4c, we observed that all AEFMs corresponded to well-established metabolic subsystems. The first, third, and fifth highest ranked AEFMs corresponded to metabolites and reactions involved in the malate-aspartate shuttle. The fourth largest AEFM closely resembled canonical TCA cycle activity, albeit with TCA intermediates citrate, isocitrate, and alpha-ketoglutarate localized to the cytosol rather than mitochondria. This AEFM also involved additional reactions transforming alpha-ketoglutarate into glutamate before remodelling into succinyl-CoA in the mitochondria. We noticed that the second largest AEFM cycled carbon mass between TCA intermediates citrate, oxaloacetate, and malate between the cytosol and mitochondria. Interestingly, this metabolic pathway corresponds to the recently discovered non-canonical TCA activity described by Arnold et al, who validated this pathway in non-small cell lung cancer, mouse embryonic, and C2C12 mouse myoblast cells by stable isotope tracing analysis (Arnold et al, 2022). This non-canonical TCA activity was further observed in glutamine carbons 3 and 4 in AEFM 2 (Figure 4e). We also found AEFMs 1, 3, and 5 in Figure 4d-e corresponded to the malate-asparate shuttle. Only the terminal glutamine carbon in Figure 4e contained source-to-sink AEFMs that decarboxylated glutamine-derived alpha-ketoglutarate into carbon dioxide.

Our analyses revealed that the most active glutamine-derived carbon AEFM weights corresponded to well-known metabolic subsystems. This finding led us to ask whether we could summarize our AEFM analysis as a simplified metabolic model describing the majority of glutamine carbon remodelling in the HepG2 flux network. The resulting schematic is shown in Figure 4f which consists of the following three major metabolic pathways: canonical TCA cycle, non-canonical TCA cycle, and the malate-aspartate shuttle. These metabolic

pathways were chosen based on the number of top glutamine-derived carbon AEFMs that overlapped with reactions within these subsystems. Our results suggest input glutamine is converted into cytosolic glutamate which is transported into the mitochondria through aspartate/glutamate antiporters encoded by *SLC25A12* and *SLC25A13* annotated in the HepG2 metabolic model. Mitochondrial glutamate is then converted into TCA intermediate alpha-ketoglutarate and citrate. Instead of following the canonical reactions of the TCA cycle, the mitochondrial citrate is exported to the cytosol where it can participate in non-canonical TCA cycle activity or the malate-aspartate shuttle to re-enter the mitochondria. Taken together, the dominant carbon AEFMs predicted by our model suggest HepG2 glutamine is conserved within TCA intermediates and regenerated through subsequent reactions involving cytosolic citrate.

Discussion

No! Too offensive!

Since the proposal of molecular EFMs, much effort has been made t compose steady state metabolic fluxes onto these pathways with little success. By defining EFMs with respect to individual atoms rather than molecules, we introduced the concept of AEFMs which describe how individual atoms are remodelled within a metabolic network. As AEFMs never involv S ways, they are easier to interpret than their molecular EFM counterparts which may involve multiple inputs and outputs to balance molecular stoichiometries. Using the atom mapping program RXNMapper, we adapted our previously described CHMC method to efficiently enumerate carbon and nitrogen AEFMs in five GEMs. Our structural analysis systematically characterized source metabolite remodelling

at the atomic level. By comparing the number of [illegible] across atoms within the same source metabolite, we revealed how certain a[illegible] markable variability in the number of pathways transited by their c[illegible] analysis, in particular, may be useful for designing non-uniformly l[illegible] ope tracers to identify fluxes within specific metabolic subsystems. Our final AEFM weight analysis of the HepG2 network revealed that the majority of glutamine carbons were remodelled via cyclic pathways regenerating TCA intermediates through cytosolic citrate. Excitingly, our model also predicted non-canonical TCA activity which has not been previously described in HepG2 cells.

kill "final"

As a systematic method to explain the flow of metabolites, the problem of molecular EFM enumeration has historically limited their applicability to small-scale networks. Many algorithmic advances enumeration have incrementally improved molecular EFM enumeration, yet never feasibly beyond networks with > 100 metabolites and reactions (Kamp & Schuster, 2006; Terzer & Stelling, 2008; van Klinken & Willems van Dijk, 2016). Molecular EFM enumeration is generally regarded as unfeasible for large-scale networks—so much so that numerous methods have been developed to enumerate a subset of molecular EFMs constrained by pathway length (de Figueiredo et al, 2009), reaction thermodynamics (Gerstl et al, 2015), participating metabolites (Pey & Planes, 2014), participating reactions (Kaleta et al, 2009), gene regulatory rules (Jungreuthmayer et al, 2013), metabolic subsystems (Schuster et al, 2002; Schwartz et al, 2007), extracellular flux measurements (Soons et al, 2011; Jungers et al, 2011), or even random sampling (Machado et al, 2012). While our HepG2 results did show the majority of source metabolite fluxes were explained by a small number of AEFMs, contextualizing the significance of these dominant pathway fluxes is only possible when all AEFMs are known. Furthermore, reaction thermodynamics, gene regulatory rules,

Would have found some of our results, but not non-canonical TCA?

and other constraints may chang suggest that enumerating the short pathways would fail to recover the majority of TCA pathways, as the top five ranked AEFMs across all glutamine carbons averaged 6 reactions in length. By computing AEFMs, instead, we could feasibly enumerate atomic pathways in genome-scale networks containing hundreds of metabolites and reactions within minutes or hours on consumer computing hardware. We are hopeful that algorithmic tricks borrowed from molecular EFM enumeration tools, such as linear pathway compression, could further decrease memory usage and run times.

Aside from carbon and nitrogen AEFMs, our work is broadly applicable to other types of chemical elements. One could feasibly enumerate oxygen or hydrogen EFMs in small-scale networks, although the significance of these pathways is unclear for biochemical networks given the ubiquitous nature of these atoms. We also caution against enumerating hydrogen AEFMs, in particular, given limitations associated with RXNMapper. Unless explicitly represented in the reaction SMILES string, RXNMapper does not assign hydrogen atom mappings. While this may seem trivial, RXNMapper cannot map atoms in reaction SMILES strings longer th d could also be applied to study sulphur metabolism delling, or phosphorus metabolism to quantify n ochemist to any perio for st networks.

The AEFM framework opens many new potential computational / analytical problems. For instance, ...

iWe analyzed one set of fluxes for HepG2 cells. If one had fluxes under differential conditions one could envision differential AEFM analysis which may reveal....

While statistical analyses, such as different AEFM weigh changes in individual pathways altered across condition metabolic pathway may impact weights in multiple AEFMs. For example, the majority

converge on
common
metabolites

of AEFMs across distinct source metabolites are not "molecularly" independent. By this, we imply that some AEFMs may correspond to the same sequence of metabolites or different metabolite/atom sequences stemming from shared reactions. Their weights will likewise be related through these common reactions. This problem may perhaps be addressed by formalizing a relationship between atomic and molecular EFMs. While we have taken significant steps towards solving the AEFM enumeration and flux decomposition problem, we expect rapid computational interest analyzing AEFM weights in the near future.

Materials and Methods

Reagents and Tools table

See tables/reagents-and-tools-table.xls.

Methods and Protocols

Datasets

Five highly-curated GEMs were chosen in this study to reflect a diversity of organisms, metabolites and biochemical reactions. Four networks were obtained from the BiGG UCSD database (King et al, 2016) in ascending order of network size. Datasets were downloaded as SBML (.xml) files, and the corresponding stoichiometry matrices, metabolites and reactions were extracted using the Julia SBML.jl package. The HepG2 liver cancer cell line dataset was chosen for its single set of estimated steady state fluxes estimated from a combination of protein and metabolite abundance data. These cells were grown in DMEM with 10% fetal

calf serum, 22mM glucose, and 1.8mM glutamine (Nilsson et al, 2020b). The steady state fluxes were recomputed from the flux balance analysis (FBA) model provided by the authors (figure3_metabolitebalances.m; MATLAB R2018a) (Nilsson et al 2020b; Data ref: Nilsson et al 2020a). Across all models, we modelled all reversible reactions as net forward reactions to simplify the number of molecular and AEFMs as the majority of experimental and computational methods can only estimate net fluxes. We assumed the reaction stoichiometries of the BiGG datasets represented the canonical net directions of all reversible reaction. For the HepG2 dataset, the net directions were chosen based on the net fluxes estimated from the FBA model.

Extracting metabolites SMILES strings

SMILES structures were not encoded in the BiGG SBML files nor the HepG2 dataset. Thus, we manually extracted SMILES strings corresponding to each metabolite across all GEMs. Isomeric SMILES strings were taken from PubChem (Kim et al, 2023), the Human Metabolome Database (Wishart et al, 2022), or MetaNetX database (Moretti et al, 2021). Metabolites shared across GEMs were assigned the same SMILES structure in addition to identical metabolites stratified across distinct subcellular compartments. Generic structures such as R-groups were not modelled in this study and were replaced with a single bonded hydrogen under the assumption this modification would not alter atom mappings, especially since RXNMapper does not map hydrogen atoms unless they are explicitly represented in the SMILES strings. SMILES strings were also not assigned to metabolites with ambiguous structures nor pseudometabolites with no known chemical structure. All pseudometabolites and reactions involving pseudometabolites were removed from the stoichiometry matrix

(Table 2).

Constructing reaction SMILES strings

RXNMapper performs atom mapping on a string that encodes the SMILES structures of all substrates and products for a given reaction. These reaction SMILES strings are constructed by concatenating SMILES strings of individual substrates and products. Multiple substrates or products are separated by the symbol '.', while the reaction arrow delimiting substrates and products is denoted by the symbol '>>'. The substrate and product SMILES strings are concatenated in order of metabolite appearance within the stoichiometry matrix, and multiple stoichiometric copies are consecutively repeated in the reaction SMILES strings. Non-integer stoichiometric coefficients are therefore not allowed and these pseudoreactions were removed from the stoichiometry matrix (Table 2). To maintain steady state flux in the HepG2 network, flux along pseudoreactions were replaced with unimolecular source and sink fluxes consuming each pseudoreaction substrate and producing each pseudoreaction product. We manually checked several reactions to ensure the assigned atom mappings were correct. Crucially, we found RXNMapper to fail on nearly all classes of transaminase reactions that were not ATP-dependent. This problem has previously been reported in an atom mapping benchmark study prior to the publication of RXNMapper (Preciat Gonzalez et al, 2017). These transaminase reactions were identified in all metabolic networks and manually atom mapped based on the KEGG reaction mappings (Figure EV6).

Pre-processing genome-scale metabolic models

Several pre-processing steps were required to compute AEFMs from the GEMs. The stoichiometric coefficients of external reactions were set to one and their fluxes rescaled to carry equivalent units of source and sink metabolites. All reactions containing pseudometabolites with no known SMILES strings were removed from the stoichiometry matrix and replaced with unimolecular source and sink reactions carrying equivalent units of flux. The same procedure was also applied to reactions containing non-integer stoichiometries. These pseudometabolites were next removed from the stoichiometry matrix rows, and unimolecular reactions involving the same external or internal metabolites were aggregated. Finally, we had to remove a small number of reactions whose reaction SMILES strings exceeded the 512 token limit of RXNMapper. These reactions were likewise converted into unimolecular source and sink reactions to maintain steady state flux across the network. All pre-processing steps described above are wrapped into convenience functions within our Julia package MarkovWeightedEFMs.jl.

Benchmarking enumeration of atomic-versus-standard EFM

We benchmarked our Julia package MarkovWeightedEFMs.jl version 2.0 running Julia version 1.10.2 to enumerate AEFMs versus the molecular EFM enumeration program FluxModeCalculator (van Klinken & Willems van Dijk (2016)) running MATLAB version R2018a with default parameters including network compression. We chose FluxModeCalculator because it is the state-of-the-art program for enumerating molecular EFMs. Benchmarks were performed on a 64-bit Linux operating system with 64GB of memory and an AMD 5950X

CPU with 16 cores/32 threads. All benchmarks reported the wall time to run the atomic or molecular EFM enumeration functions and did not include startup times to open Julia/MATLAB, precompile Julia functions, or load packages and input data. A benchmark time of 'did not finish' (DNF) was assigned if either program could not complete (A)EFM enumeration within 7 days. In practice, only FluxModeCalculator failed to enumerate the molecular EFMs in three of the five datasets.

Constructing atomic transition graphs

For a user-specified atom within a given source metabolite, we identify all possible metabolite/atom states that are traversable by that initial state and constrained by the reaction atom mapping predictions. For reactions involving single stoichiometric copies of each substrate and product, the atom of interest will always move from a single substrate to product molecule. If this atom is present in a substrate with multiple stoichiometric copies, we assume there is an equal probability of that atom occurring in either copy. Depending on the substrate stoichiometric copy, that atom may move to (i) different product metabolite/atom states, (ii) the same product but at different atomic positions, or (iii) the same metabolite/atom position. In all cases, we set the atomic flux of the atom transitioning from either stoichiometric copy equal to the reaction fluxes to maintain atomic mass conservation (Figure EV 5). The resulting atomic transition graph is completed by connecting any sink fluxes to the root of the tree initialized on the source metabolite/atom position.

ACHMC model

Following Chitpin & Perkins, the atomic transition graphs were analyzed by an atomic version of our CHMC (ACHMC) method to enumerate the corresponding AEFMs and compute their weights. In this formulation, each ACHMC state corresponds to a sequence of metabolite-atom positions transited by a given source metabolite atom. Each simple cycled identified by the ACHMC corresponds to an AEFM rooted on a user-specified source metabolite atom. For the HepG2 flux network, we computed the probabilities of each AEFM by steady state analysis of the ACHMC. These AEFM probabilities were scaled to weights such that the totality of source-to-sink AEFMs was equal to the unimolecular input flux producing that source metabolite.

Data availability

Our software package MarkovWeightedEFMs.jl is available at `https://github.com/jchitpin/MarkovWeightedEFMs.jl`. All source codes to reproduce this study are available at `https://www.github.com/jchitpin/reproduce-efm-paper-2024`.

Acknowledgements

This research was supported by the Natural Sciences and Engineering Research Council of Canada (NSERC), Discovery grant RGPIN/06604-2019 and a Digital Research Alliance of Canada (DRAC) Resources for Research Groups grant to TJP (RRG-TPERKINS). J.G.C. was supported by an NSERC CREATE Matrix Metabolomics Scholarship, NSERC Alexan-

der Graham Bell Canada Graduate Scholarship, and Ontario Graduate Scholarship.

Author contributions

Justin G. Chitpin: Conceptualization; data curation; methodology; software; formal analysis; investigation; visualization; writing - original draft; writing - review and editing.

Theodore J. Perkins: Conceptualization; methodology; supervision; funding acquisition; investigation; project administration; writing - review and editing.

Disclosure and competing interests statement

The authors declare no competing interests.

References

Acuña V, Chierichetti F, Lacroix V, Marchetti-Spaccamela A, Sagot MF, Stougie L (2009) Modes and cuts in metabolic networks: Complexity and algorithms. *BioSystems* 95: 51–60

Ahn E, Kumar P, Mukha D, Tzur A, Shlomi T (2017) Temporal fluxomics reveals oscillations in TCA cycle flux throughout the mammalian cell cycle. *Mol Syst Biol* 13: 953

Antoniewicz MR (2018) A guide to 13C metabolic flux analysis for the cancer biologist. *Exp Mol Med* 50: 1

Arnold PK, et al (2022) A non-canonical tricarboxylic acid cycle underlies cellular identity. *Nature* 603: 477–481

Basan M, Hui S, Okano H, Zhang Z, Shen Y, Williamson JR, Hwa T (2015) Overflow metabolism in escherichia coli results from efficient proteome allocation. *Nature* 528: 99–104

Becker SA, Palsson BØ (2005) BiGG http://bigg.ucsd.edu/models/isb619. [DATASET]

Bordbar A, Jamshidi N, Palsson BØ (2011) BiGG (http://bigg.ucsd.edu/models/iab_rbc_283). [DATASET]

Brunk E, et al (2018) Recon3d enables a three-dimensional view of gene variation in human metabolism. *Nat Biotechnol* 36: 272–281

Buchner BA, Zanghellini J (2021) EFMlrs: a python package for elementary flux mode enumeration via lexicographic reverse search. *BMC Bioinformatics* 22: 1–547

Buescher JM, et al (2015) A roadmap for interpreting 13C metabolite labeling patterns from cells. *Curr Opin Biotechnol* 34: 189–201

Cherkaoui S, et al (2022) A functional analysis of 180 cancer cell lines reveals conserved intrinsic metabolic programs. *Mol Syst Biol* 18: e11033

Chitpin JG, Perkins TJ (2023) A Markov constraint to uniquely identify elementary flux mode weights in unimolecular metabolic networks. *J Theor Biol* 575: 111632–111632

de Figueiredo LF, Podhorski A, Rubio A, Kaleta C, Beasley JE, Schuster S, Planes FJ

(2009) Computing the shortest elementary flux modes in genome-scale metabolic networks. *Bioinformatics* 25: 3158–3165

Duan L, Cooper DE, Scheidemantle G, Locasale JW, Kirsch DG, Liu X (2022) 13C tracer analysis suggests extensive recycling of endogenous CO2 in vivo. *Cancer Metab* 10: 1–11

Emwas AH, Szczepski K, Al-Younis I, Lachowicz JI, Jaremko M (2022) Fluxomics - new metabolomics approaches to monitor metabolic pathways. *Front Pharmacol* 13: 805782–805782

Endo Y, Kanno T, Nakajima T (2022) Fatty acid metabolism in T-cell function and differentiation. *Int Immunol* 34: 579–587

Fakih I, Got J, Robles-Rodriguez CE, Siegel A, Forano E, Muñoz-Tamayo R (2023) Dynamic genome-based metabolic modeling of the predominant cellulolytic rumen bacterium fibrobacter succinogenes s85. *mSystems* 8: e0102722–e0102722

Franchina DG, et al (2022) Glutathione-dependent redox balance characterizes the distinct metabolic properties of follicular and marginal zone B cells. *Nat Commun* 13: 1789–1789

Gagneur J, Klamt S (2004) Computation of elementary modes: a unifying framework and the new binary approach. *BMC Bioinformatics* 5: 175–175

Gelbach PE, Cetin H, Finley SD (2024) Flux sampling in genome-scale metabolic modeling of microbial communities. *BMC Bioinformatics* 25: 45

Gerstl MP, Ruckerbauer DE, Mattanovich D, Jungreuthmayer C, Zanghellini J (2015)

Metabolomics integrated elementary flux mode analysis in large metabolic networks. *Sci Rep* 5: 8930

González-Arrué N, Inostroza I, Conejeros R, Rivas-Astroza M (2023) Phenotype-specific estimation of metabolic fluxes using gene expression data. *iScience* 26: 106201–106201

Gustafsson J, Roshanzamir F, Hagnestål A, Patel SM, Daudu OI, Becker DF, Robinson JL, Nielsen J (2024) Metabolic collaboration between cells in the tumor microenvironment has a negligible effect on tumor growth. *Innovation* 5: 100583

Huang Y, Mohanty V, Dede M, Tsai K, Daher M, Li L, Rezvani K, Chen K (2023) Characterizing cancer metabolism from bulk and single-cell rna-seq data using METAFlux. *Nat Commun* 14: 4883–4883

Jackson BT, Finley LWS (2024) Metabolic regulation of the hallmarks of stem cell biology. *Cell Stem Cell* 31: 161–180

Jungers RM, Zamorano F, Blondel VD, Wouwer AV, Bastin G (2011) Fast computation of minimal elementary decompositions of metabolic flux vectors. *Automatica* 47: 1255–1259

Jungreuthmayer C, Ruckerbauer DE, Zanghellini J (2013) regEfmtool: Speeding up elementary flux mode calculation using transcriptional regulatory rules in the form of three-state logic. *BioSystems* 113: 37–39

Kaleta C, de Figueiredo LF, Schuster S (2009) Can the whole be less than the sum of its parts? pathway analysis in genome-scale metabolic networks using elementary flux patterns. *Genome Res* 19: 1872–1883

Kamp Av, Schuster S (2006) Metatool 5.0: fast and flexible elementary modes analysis. *Bioinformatics* 22: 1930–1931

Kaste JAM, Shachar-Hill Y (2023) Accurate flux predictions using tissue-specific gene expression in plant metabolic modeling. *Bioinformatics* 39

Kim S, et al (2023) PubChem 2023 update. *Nucleic Acids Res* 51: D1373–D1380

King ZA, Lu J, Dräger A, Miller P, Federowicz S, Lerman JA, Ebrahim A, Palsson BØ, Lewis NE (2016) BiGG Models: a platform for integrating, standardizing and sharing genome-scale models. *Nucleic Acids Res* 44: D515–D522

Kochanowski K, Okano H, Patsalo V, Williamson J, Sauer U, Hwa T (2021) Global coordination of metabolic pathways in ¡i¿escherichia coli¡/i¿ by active and passive regulation. *Mol Syst Biol* 17: e10064

Lee JY, Han Y, Styczynski MP (2023) Towards inferring absolute concentrations from relative abundance in time-course gc-ms metabolomics data. *Mol Omics* 19: 126–136

Lee WD, Mukha D, Aizenshtein E, Shlomi T (2019) Spatial-fluxomics provides a subcellular-compartmentalized view of reductive glutamine metabolism in cancer cells. *Nat Commun* 10: 1351–1351

Machado D, Soons Z, Patil KR, Ferreira EC, Rocha I (2012) Random sampling of elementary flux modes in large-scale metabolic networks. *Bioinformatics* 28: i515–i521

Matsuda F, Furusawa C, Kondo T, Ishii J, Shimizu H, Kondo A (2011) Engineering strategy of yeast metabolism for higher alcohol production. *Microb Cell Fact* 10: 70–70

McKinlay JB, Harwood CS (2010) Carbon dioxide fixation as a central redox cofactor recycling mechanism in bacteria. *Proc Natl Acad Sci U S A* 107: 11669–11675

Moretti S, Tran V, Mehl F, Ibberson M, Pagni M (2021) MetaNetX/MNXref: unified namespace for metabolites and biochemical reactions in the context of metabolic models. *Nucleic Acids Res* 49: D570–D574

Mpabanzi L, et al (2022) Fluxomics reveals cellular and molecular basis of increased renal ammoniagenesis. *NPJ Syst Biol Appl* 8: 49–10

Nilsson A, Haanstra JR, Engqvist M, Gerding A, Bakker BM, Klingmüller U, Teusink B, Nielsen J (2020a) GitHub (https://github.com/sysbiochalmers/livercellmetabolismsimulation). [DATASET]

Nilsson A, Haanstra JR, Engqvist M, Gerding A, Bakker BM, Klingmüller U, Teusink B, Nielsen J (2020b) Quantitative analysis of amino acid metabolism in liver cancer links glutamate excretion to nucleotide synthesis. *Proc Natl Acad Sci U S A* 117: 10294–10304

Nobile MS, Coelho V, Pescini D, Damiani C (2021) Accelerated global sensitivity analysis of genome-wide constraint-based metabolic models. *BMC Bioinformatics* 22: 78–78

Orth JD, Fleming RMT, Palsson BØ (2010a) BiGG (http://bigg.ucsd.edu/models/e_coli_core). [DATASET]

Orth JD, Fleming RMT, Palsson BØ (2010b) Reconstruction and use of microbial metabolic networks: the core Escherichia coli metabolic model as an educational guide. *Ecosal Plus* 4

Pey J, Planes FJ (2014) Direct calculation of elementary flux modes satisfying several biological constraints in genome-scale metabolic networks. *Bioinformatics* 30: 2197–2203

Preciat Gonzalez GA, El Assal LRP, Noronha A, Thiele I, Haraldsdóttir HS, Fleming RMT (2017) Comparative evaluation of atom mapping algorithms for balanced metabolic reactions: application to recon 3d. *J Cheminform* 9: 39–39

Robinson JL, et al (2020) An atlas of human metabolism. *Sci Signal* 13

Sanford K, Soucaille P, Whited G, Chotani G (2002) Genomics to fluxomics and physiomics – pathway engineering. *Curr Opin Microbiol* 5: 318–322

Schuetz R, Zamboni N, Zampieri M, Heinemann M, Sauer U (2012) Multidimensional optimality of microbial metabolism. *Science* 336: 601–604

Schuster S, Dandekar T, Fell DA (1999) Detection of elementary flux modes in biochemical networks: a promising tool for pathway analysis and metabolic engineering. *Trends Biotechnol* 17: 53–60

Schuster S, Hilgetag C (1994) On elementary flux modes in biochemical reaction systems at steady state. *J Biol Syst* 2: 165–182

Schuster S, Pfeiffer T, Moldenhauer F, Koch I, Dandekar T (2002) Exploring the pathway structure of metabolism: decomposition into subnetworks and application to mycoplasma pneumoniae. *Bioinformatics* 18: 351–361

Schwaller P, Hoover B, Reymond JL, Strobelt H, Laino T (2021) Extraction of organic chemistry grammar from unsupervised learning of chemical reactions. *Sci Adv* 7: eabe4166

Schwartz JM, Gaugain C, Nacher JC, de Daruvar A, Kanehisa M (2007) Observing metabolic functions at the genome scale. *Genome Biol* 8: R123–R123

Schwartz JM, Kanehisa M (2006) Quantitative elementary mode analysis of metabolic pathways: the example of yeast glycolysis. *BMC Bioinformatics* 7: 186–186

Soons ZI, Ferreira EC, Rocha I (2011) Identification of minimal metabolic pathway models consistent with phenotypic data. *J Process Control* 21: 1483–1492

Terzer M, Stelling J (2008) Large-scale computation of elementary flux modes with bit pattern trees. *Bioinformatics* 24: 2229–2235

Thiele I, Vo TD, Price ND, Palsson BØ (2005) BiGG (http://bigg.ucsd.edu/models/iit341). [DATASET]

Toya Y, Shimizu H (2022) Metabolic pathway engineering for the non-growth-associated succinate production in escherichia coli based on flux solution space. *J Biosci Bioeng* 134: 29–33

Uematsu S, et al (2022) Multi-omics-based label-free metabolic flux inference reveals obesity-associated dysregulatory mechanisms in liver glucose metabolism. *iScience* 25: 103787–103787

van Klinken JB, Willems van Dijk K (2016) FluxModeCalculator: an efficient tool for large-scale flux mode computation. *Bioinformatics* 32: 1265–1266

Vander Heiden MG, Cantley LC, Thompson CB (2009) Understanding the warburg effect: The metabolic requirements of cell proliferation. *Science* 324: 1029–1033

Wang H, et al (2021) Genome-scale metabolic network reconstruction of model animals as a platform for translational research. *Proc Natl Acad Sci U S A* 118

Wang L, et al (2022a) Spatially resolved isotope tracing reveals tissue metabolic activity. *Nat Methods* 19: 223–230

Wang R, et al (2022b) Global stable-isotope tracing metabolomics reveals system-wide metabolic alternations in aging Drosophila. *Nat Commun* 13: 3518–3518

Wishart DS, et al (2022) HMDB 5.0: the Human Metabolome Database for 2022. *Nucleic Acids Res* 50: D622–D631

Wu C, Herold RA, Knoshaug EP, Wang B, Xiong W, Laurens LML (2019) Fluxomic analysis reveals central carbon metabolism adaptation for diazotroph Azotobacter vinelandii ammonium excretion. *Sci Rep* 9: 13209–11

Ye Y, Yu B, Wang H, Yi F (2023) Glutamine metabolic reprogramming in hepatocellular carcinoma. *Front Mol Biosci* 10: 1242059–1242059

Yoo HC, Yu YC, Sung Y, Han JM (2020) Glutamine reliance in cell metabolism. *Exp Mol Med* 52: 1496–1516

Yu JSL, et al (2022) Microbial communities form rich extracellular metabolomes that foster metabolic interactions and promote drug tolerance. *Nat Microbiol* 7: 542–555

Zamboni N, Saghatelian A, Patti G (2015) Defining the metabolome: Size, flux, and regulation. *Mol Cell* 58: 699–706

Zampieri M, Hörl M, Hotz F, Müller NF, Sauer U (2019) Regulatory mechanisms underlying coordination of amino acid and glucose catabolism in escherichia coli. *Nat Commun* 10: 3354

Figure legends

Fig 1: Atomic cycle-history Markov chain (ACHMC) pipeline. (a) An example molecular EFM in glycolysis. (b) An example AEFM tracing the flow of a given glucose carbon in glycolysis. (c) The major steps within our pipeline to enumerate AEFMs and compute their weights.

Fig 2: AEFM enumeration is computationally tractable compared to molecular EFMs in five GEMs. (a) Number of atomic and molecular EFMs in the five GEMs. (b) The run time of our MarkovWeightedEFMs.jl method versus FluxModeCalculator running in serial. (c) Serial run time scaling of MarkovWeightedEFMs.jl as a function of the number of AEFMs. (d) Speedup of selected ACHMCs as a functional of additional threads.

Fig 3: Structural analysis of AEFMs in five genome-scale metabolic models. (a-b) Categories of carbon and nitrogen EFMs. (c-d) Distribution carbon and nitrogen AEFM lengths. (e-f) Ratio of the greatest and smallest number of AEFMs between carbons and nitrogens within the same source metabolite. Representative metabolites from each dataset are labelled; labels with asterisks indicate that metabolite is also present in other networks.

Fig 4: Carbon AEFM weight analysis of the HepG2 metabolic flux network. (a) A small number of highly active AEFMs explain the majority of glutamine flow. (b) Subnetwork of the top five AEFMs across each glutamine carbon collapsed onto metabolites. (c-e) Sankey

diagrams of the top five glutamine carbon AEFMs. (f) Schematic mapping AEFMs onto traditionally defined metabolic pathways/subsystems. All metabolite names are labelled following the BiGG database nomenclature.

Tables and their legends

Table 1: Summary statistics for the five metabolic networks in this study.

Table 2: Number of metabolites and reactions removed from each network during pre-processing.

Expanded View Figure legends

Expanded View 1: Network visualization of the five GEMs after pre-processing. Metabolites (non-grey nodes) are connected to reaction entities (grey nodes) with node sizes proportional to their degree. Graphs were constructed in Gephi version 0.10 using the Yifan-Hu proportional network layout with default parameters. (a) E. coli core. (b) iAB RBC 283. (c) iIT341. (d) iSB619. (e) HepG2.

Expanded View 2: FluxModeCalculator scaling across multiple threads.

Expanded View 3: Most ACHMCs traverse less than 10% of the total metabolite-carbon/nitrogen state space. Each point corresponds to an ACHMC rooted on a given source metabolite carbon or nitrogen.

Expanded View 4: Cumulative explained carbon mass flow of input glucose and amino acid carbons from the HepG2 model. The cumulative explained mass flow is truncated at 99%

to avoid displaying AEFMs that carry little atomic mass flow.

Expanded View 5: Rules for assigning atomic fluxes from atom-mapped reactions with biological examples. Schematic node colours represent ground truth atom mappings across substrate(s) and product(s). Node labels correspond to distinct atomic indices of substrate and product stoichiometric copies. (a) For reactions with single stoichiometric copies of substrates and products, each atom within a given substrate moves to a single atomic position within a product with atomic flux equal to the reaction flux. (b-d) For reactions with s stoichiometric copies of a given substrate, there are s copies of each metabolite-atom state on the substrate and product side of the reaction. The probability of any atom within a substrate stoichiometric copy moving to a product metabolite-atom state remains equal to the reaction flux. (b) The stoichiometric copies of the substrate fuse together into a single product. (c) The stoichiometric copies of the substrate form distinct products. (d) The s stoichiometric copies of the substrate form s copies of products with identical atomic backbones. (e) Biological example of rule a where the atoms in each substrate map to a single position on the corresponding product. (f) Biological example of rule b where two stoichiometric copies of glutathione fuse together to form a single stoichiometric copy of glutathione disulfide. (g) Biological example of rule c where two stoichiometric copies of acetyl-CoA undergo a reaction to produce a stoichiometric copy of two distinct products.

Expanded View 6: Examples of incorrectly atom-mapped transaminase reactions present in one or more of the five networks. (a) Phenylalanine transaminase. (b) Isoleucine transaminase. (c) Tyrosine transaminase. (d) Aspartate transaminase.